

CustomerCluster: An Unsupervised Machine Learning Approach for Intelligent Customer Clustering and Behavioral Analysis

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Abstract—Customer segmentation plays a vital role in helping businesses understand customer behavior, improve marketing strategies, increase customer satisfaction, and enhance decision-making in highly competitive retail and e-commerce environments. However, traditional segmentation techniques struggle to identify meaningful customer groups due to the complex and overlapping nature of modern customer behavioral data. In this paper we propose an efficient framework for intelligent customer segmentation using four real-world datasets: Online Retail Customer Dataset, Customer Purchase Behavior (E-Commerce) Dataset, Consumer Shopping Trends Dataset, and Customer Purchases Behaviour Dataset. The proposed framework applies multiple unsupervised machine learning algorithms including K-Means, Agglomerative Clustering, DBSCAN, Mean Shift, Gaussian Mixture Model (GMM), and OPTICS after performing preprocessing, feature engineering, normalization, encoding, and feature scaling. Experimental results demonstrated that K-Means consistently achieved the best overall clustering performance across most datasets, obtaining the highest Silhouette Scores and producing clear and interpretable customer groups. The findings confirm that centroid-based clustering approaches are more effective than density-based methods for customer behavioral analysis, enabling businesses to generate actionable insights, improve customer targeting, and support personalized marketing strategies.

I. INTRODUCTION

In today's highly competitive market, understanding customer behavior has become one of the most important factors for a business to keep a sustainable competitive advantage. Customer segmentation helps organizations optimize their marketing campaigns, increase customer satisfaction, and improve resource allocation. Research has shown that marketing to specific customer segments results in up to 30% higher conversion rates and much better customer retention. As the number of customers and the complexity of behavioral patterns grow, extracting actionable insights from such data is no longer a competitive advantage, but a baseline requirement for any customer-centric business.

However the availability of many customer data, organizations are still struggling to identify meaningful customer segments due to the complex nature of behavioral data. Traditional

segmentation methods typically based on manual observation and basic statistical techniques, which fail to capture the complex, multi-dimensional patterns in modern purchasing behavior. Customer attributes have non-linear interactions over demographic, financial, and behavioral factors, making them difficult to model using simple approaches. The increase of customer information continues to growth, it becomes challenging to develop accurate marketing strategies using manual methods. Furthermore, large datasets often contain noisy or inconsistent data that can negatively affect the performance of algorithms. Therefore, there is a critical need for an automated, data-driven approach capable of partitioning a diversity of customers into meaningful and actionable clusters. solving this challenge allows businesses to move from generic marketing strategies to highly personalized and optimized customer experiences.

Machine learning (ML) is a subfield of artificial intelligence that focuses on designing algorithms that can learn patterns and predict from data without being directly programmed. ML methods are divided into supervised learning, which trains models on labeled data; unsupervised learning, which discovers hidden structures in unlabeled data; and reinforcement learning, which optimizes sequential decision-making through trial and error. Using advanced algorithms, machine learning models can autonomously learn from data, identify complex patterns, and make data-driven decisions with minimal human intervention. The ability of these models to continuously improve their performance and effectively manage complex data makes them precious for modern analytical tasks.

To solve the complexities of customer segmentation, unsupervised machine learning techniques, especially clustering algorithms, offer a highly effective solution. Clustering algorithms such as K-Means, DBSCAN, Mean Shift, Gaussian Mixture, OPTICS, and Agglomerative Hierarchical Clustering have been widely applied to customer datasets. These algorithms demonstrate powerful performance in discovering actionable behavioral patterns, enabling businesses to divide customers into meaningful and identical groups. Furthermore, internal evaluation metrics such as the Silhouette Score and

Davies–Bouldin Index provide label-free methods to assess cluster quality and determine the optimal number of segments. By applying these machine learning methods, it is possible to analyze the strengths and limitations of each algorithm under the same conditions, delivering a more accurate and scalable approach compared to traditional methods. The resulting clusters help businesses improve customer retention, increase sales, and deliver enhanced personalized customer experiences.

As a summary, our contributions are as follows:

- Applying multiple clustering algorithms to identify the most effective approach for grouping customers based on their purchasing patterns and engagement levels.
- Evaluating six unsupervised clustering algorithms — KMeans, DBSCAN, Agglomerative Clustering, Gaussian Mixture Models, MeanShift, and OPTICS, using Silhouette Score and Davies-Bouldin Index to identify the optimal model.
- Analyzing the resulting customer segments to extract actionable business insights that can directly inform targeted marketing and personalization strategies.
- Providing actionable business insights derived from the unique characteristics of each identified cluster to aid in targeted marketing and retention strategies.
- Delivering a deployable segmentation model saved with its corresponding scaler and feature configuration, enabling prediction on unseen customer data without re-training.

The rest of the paper is organized as follows: Section II reviews the related work in the field of customer segmentation. Section III presents the methodology, including the dataset, preprocessing steps, and clustering algorithms. Section IV discusses the results and evaluation metrics. Finally, Section V concludes the paper and outlines potential future work.

II. RELATED WORK

In the paper [1], Customer segmentation is one of the vital practices in CRM, categorizing individuals into groups upon age and purchasing behaviours by optimizing a unique approach for each segment to elevate business value. In this study a Hierarchical Agglomerative Clustering (HAC) algorithm implemented in R programming language will be used to analyze credit card datasets. Ultimately, data-driven clustering enables businesses to identify distinct consumer profiles that allow them to develop nuanced and more targeted marketing strategies.

The paper [2] uses K-Means on RFM-like features extracted from transaction data. The authors use both standard and Mini-Batch K-Means, evaluate clusters using Silhouette Score and Elbow method, and aim to generate actionable segments (e.g. high-value vs low-value customers) for marketing. Limitations include sensitivity to initialization, loss of temporal patterns in aggregate features, and limited

comparison to other algorithms.

In [3] Description: This paper provides a comparative analysis of five machine learning clustering algorithms—K-Means, BIRCH, Agglomerative Hierarchical Clustering, DBSCAN, and OPTICS—spanning centroid-based, connectivity-based, and density-based approaches. Utilizing an open-source dataset of hotel customers, the study evaluates how effectively these techniques handle mixed data structures to segment market audiences and enhance corporate communication strategies. Advantages: It establishes a comprehensive benchmark by assessing three distinct clustering paradigms and implements rigorous validation metrics, specifically using the Davies-Bouldin and Silhouette scores, to mathematically verify segment quality. Disadvantages: The empirical findings are heavily tied to a single open-source hotel industry dataset, meaning the performance conclusions and algorithm superiority rankings may not directly generalize to other commercial sectors. Result: The comparative evaluation reveals that Agglomerative Hierarchical Clustering outperforms the alternative algorithms on this data, while proving that integrating Gower’s distance significantly optimizes density-based and connectivity-based algorithms when handling mixed data types.

In [4], K-Means and Fuzzy C-Means are compared on RFM data in the article "RFM ranking - an effective approach to customer segmentation". It proposes a better way to choose the initial centroids in K-Means. It also highlights the trade-off between the crisp segments (K-Means) and soft overlapping memberships (Fuzzy C-Means) for better marketing and customer retention.

In the paper [5] gives an analysis to a large telecom dataset (100,000 customers, 220 features). They used dimensionality reduction (PCA vs autoencoders) prior to K-Means clustering. The quality and speed of clustering improved with the right amount of reduction. When using autoencoders it often outperforms PCA when identifying nonlinear patterns. Caveats include sensitivity to model structure and lower interpretability.

The research paper [6] utilizes a **large-scale e-commerce dataset from Pakistan**, the authors study a proprietary high-volume transaction behavior dataset provided by a large e-commerce marketplace in Pakistan. By applying an RFMT model (Recency, Frequency, Monetary, and Tenure), they compare cluster models across K-Means, Gaussian Mixture, Hierarchical, and DBSCAN algorithms. They then use a voting technique across six validity indices to determine that stable customer segments exist within the dataset. The authors conclude that for this vertical, K-Means produced the most stable clusters consistently and was able to identify relevant customer tiers such as "Champions" and "At-Risk" shoppers. Technical hurdles include skewed distributions and

noisy datasets of transaction logs. Their findings suggest that marketing efficiency can be increased by 90 percent through multi-index validation versus single solutions. Limitations may consist of time complexity due to model voting and density sensitivity.

The study [7] examines physical retail behavior using the **Mall Customers dataset** to improve localized marketing. The optimal number of clusters is determined using the Elbow method and the algorithms are then used to generate actionable customer personas. The study shows that for this particular low-dimensional data, K-Means produces more separate and reasonable clusters than the noise-sensitive DBSCAN. The segments identified are “High Income, High Spend” and “Frugal” shoppers and provide a direct map for placement of mall advertising. Potential limitations include the small size of the dataset and the reliance on only two primary features, which may oversimplify complex behaviors.

In the paper [8] the authors use a **retail transaction dataset** to evaluate customer loyalty levels. The methodology converts the raw logs into RFM scores and makes a three-way algorithmic comparison to decide the most accurate partition. The study achieved a high Silhouette score of 0.913 with K-Means, which indicates an almost perfect separation of clusters for the retail groups. The results show that K-Medoids is more robust against data outliers and K-Means is more efficient and accurate with general shopping patterns. Potential limitations are the possible loss of temporal dynamics when transactions are aggregated into static RFM scores. The research shows that K-Means based RFM is the most effective way for retailers in increasing overall customer lifetime value.

The paper [9] proposes a practical hybrid method that combines Factor Analysis of Mixed Data (FAMD) for dimensionality reduction with K-Means and Agglomerative Hierarchical Clustering. This approach effectively handles both numerical and categorical features commonly found in customer datasets. By reducing the feature space while preserving important information, the method produces more stable and interpretable customer segments. The study demonstrates that integrating FAMD with hierarchical clustering helps reveal meaningful behavioral patterns and supports better marketing decisions. However, the authors note limitations such as sensitivity to the choice of principal components and the added computational cost during the dimensionality reduction step.

The research [10], utilizes the UCI Online Retail II dataset to tackle the challenge of manual cluster selection. The authors have developed an automated K-optimization script which analyzes the data to determine the best number of segments without any human intervention. This automated K-Means approach is implemented on RFM features to build a dynamic system for e-commerce platforms. The study shows that taking human guesswork out of the equation

results in more stable clusters and enables the model to update in real-time. Limitations are the sensitivity of the optimization script to the initial distribution and scaling of the data set.

This paper [11] To solve the need for better e-commerce customer segmentation, this study evaluated five clustering algorithms using a UK-based Online Retail Dataset (541,909 records). The authors applied the RFM (Recency, Frequency, Monetary) framework to quantify customer value, and PCA for dimensionality reduction, before testing KMeans, GMM, DBSCAN, Agglomerative Clustering, and BIRCH. Ultimately, the Gaussian Mixture Model (GMM) outperformed the others with a Silhouette Score of 0.80. The study concludes that GMM combined with PCA provides the most accurate and interpretable model for retail decision-making.

This paper [12] To help businesses select the most effective behavioral customer segmentation models, this study compared four machine learning approaches: KMeans, DBSCAN, Agglomerative Clustering, and a combination of PCA and KMeans. Using a dataset based on customer age, annual income, and spending score, the models were evaluated on their ability to create distinct customer groups. finally, Agglomerative Clustering outperformed the rest, achieving a Silhouette Score of 0.6865. The authors conclude that this comparative analysis serves as a reliable guide for organizations looking to implement efficient segmentation and enhance their marketing strategies.

This paper [13] To improve marketing strategies in the fashion industry, William et al. evaluated clustering algorithms using transaction data from an Indonesian company. Following the CRISP-DM framework, they applied Standard Scaler and PCA preprocessing before comparing K-Means and DBSCAN. K-Means proved superior, identifying seven optimal customer clusters with a Silhouette Score of 0.549 and a Davies-Bouldin Index of 0.593. To bridge the gap between research and practical application, the authors successfully deployed this K-Means model into a FastAPI and React-based web dashboard to support real-time business decision-making.

This paper [14] To overcome the challenges of high-dimensional data and parameter selection in digital marketing segmentation, this study proposed a novel hybrid framework called K-means-QLDE, which combines PCA, a Q-learning-based Differential Evolution algorithm, and K-Means. Using a Kaggle retail dataset transformed by the RFM (Recency, Frequency, Monetary) model, the authors applied the elbow method to successfully group customers into six distinct behavioral clusters. The proposed approach proved highly effective, achieving over 95% classification accuracy when validated against ANN, KSVM, Decision Tree, and AdaBoost models. The researchers conclude that this enhanced framework provides a highly accurate way

to improve marketing efficiency and business decision-making.

This paper [15] addresses the limitations of traditional RFM (Recency, Frequency, Monetary) models in the dynamic e-commerce environment by proposing an extended RFM framework. The authors incorporated three additional variables — campaign share, basket depth, and inter-order interval variance — to better capture modern customer behavior, promotional responsiveness, and purchase regularity. Using real transaction data spanning 2024 to 2025, the study systematically evaluated three clustering algorithms: K-Means, K-Medoids, and Fuzzy C-Means. Model performance was rigorously assessed using multiple internal validation metrics, including the Silhouette Score, Dunn Index, and Davies–Bouldin Index. Ultimately, K-Means demonstrated the strongest overall performance, successfully identifying nine distinct and meaningful customer segments. These segments offered clearer differentiation in terms of purchasing patterns, promotional sensitivity, and behavioral loyalty compared to conventional RFM-based approaches.

This paper [16] explores advanced and more intelligent approaches to customer segmentation by moving beyond traditional methods. The study investigates multiple clustering techniques to discover more accurate, stable, and actionable customer groups based on actual shopping behavior. The authors enhanced the segmentation process by incorporating additional dynamic features, including customer growth rate, level of engagement with the brand, and the effectiveness of various business strategies applied over time. By integrating decision-making frameworks with unsupervised learning, the researchers proposed a novel hybrid clustering method named APK-Means++. This innovative approach intelligently combines the strengths of K-Means++ and Affinity Propagation algorithms. The resulting hybrid model demonstrates improved performance in identifying natural customer segments with higher cohesion and better separation compared to conventional techniques.

This paper [17] takes a very practical approach to understanding different types of retail customers. The researchers utilized a massive dataset from UK online shopping and aimed to make the segmentation process as clear, interpretable, and actionable as possible. Rather than relying solely on traditional RFM variables, they incorporated real-life behavioral factors such as holiday shopping patterns and order cancellation frequency to better reflect actual customer behavior. They comprehensively evaluated ten different clustering algorithms to determine which method produced the most meaningful customer groups. A standout contribution of this study is the use of SHAP (SHapley Additive exPlanations) values to explain the reasoning behind cluster assignments, significantly improving the transparency and trustworthiness of the results. In the end, DBSCAN emerged as the superior method, identifying clear and highly useful customer segments that outperformed other popular

techniques. DBSCAN achieved an impressive Silhouette Score of 0.986, far exceeding the 0.72 benchmark commonly reported in the literature. These findings highlight the effectiveness of density-based clustering when applied to large-scale transactional data.

This paper [18] This study looks at how retailers can better understand their customers by grouping them based on their shopping habits. Using purchase data from 2022 to 2024, the researchers combined RFM analysis (which considers how recently, how often, and how much customers spend) with the K-Means clustering method. They made sure the results were reliable by standardizing the data and carefully choosing the right number of groups. In the end, they identified five clear customer types—Platinum, Gold, Silver, Bronze, and Bad—each with unique behaviors that can help businesses design more targeted marketing, loyalty programs, and win-back campaigns. The approach was tested and validated using various techniques and visualizations, showing that it works well and can be scaled for different retail settings.

This paper [19] investigates effective customer segmentation strategies specifically within the Turkish retail market. The researchers conducted a comprehensive comparative analysis of several clustering algorithms using both behavioral and demographic customer data. Their primary objective was to identify the most suitable techniques that could translate into practical and effective marketing strategies. The study revealed that K-Means delivered highly interpretable and statistically robust results, making it particularly valuable for deriving actionable business insights. In contrast, DBSCAN excelled at detecting unusual and non-standard customer groups that traditional methods often overlook. Meanwhile, Gaussian Mixture Models (GMM) and Self-Organizing Maps (SOM) provided richer, more nuanced insights into complex customer patterns and subtle behavioral differences. Overall, the study emphasizes the importance of selecting the right clustering approach based on specific business goals in emerging retail markets.

This paper [20] presents a practical case study on improving customer understanding and marketing effectiveness for fixed deposit customers in a Portuguese bank. The authors first analyzed the limitations of existing marketing approaches and then proposed a novel hybrid clustering method called KM-DBSCAN, which intelligently integrates K-Means and DBSCAN algorithms. This new technique effectively handles different types of customer data and incorporates an automatic parameter tuning mechanism to enhance clustering quality. Compared to the standard DBSCAN algorithm, the KM-DBSCAN method achieved a 15% improvement in clustering accuracy. As a result, the bank was able to identify four distinct and meaningful customer segments, enabling the design of more personalized and targeted marketing strategies for each group.

TABLE I: The Papers Overview

Paper name	Year	Method used	Result summary	Advantage	Disadvantage
Customer Segmentation Using Hierarchical Agglomerative Clustering	2019	Agglomerative Algorithm	A reproducible, visually intuitive framework in R to segment users without supplying group numbers a priori and reach groups directly for targeting.	A reproducible, visually intuitive framework in R to segment users without supplying group numbers a priori and reach groups directly for targeting.	Poor scalability for large datasets, very sensitive to noise and generates static definitions based solely on boundaries defined by the user.
Customer Segmentation Based on RFM Model Using K-Means, K-Medoids, and DBSCAN	2020	RFM; K-Means; K-Medoids; DBSCAN	Means, K-Medoids, DBSCAN-Means achieved a very high Silhouette Score of 0.913 and performed best overall.	High cluster quality and clear comparison between algorithms	Loses time-related patterns when using static RFM scores
A Comparative Dimensionality Reduction Study in Telecom Customer Segmentation Using Deep Learning and PCA	2020	PCA; Autoencoder; K-Means	Autoencoders performed better than PCA at capturing complex patterns and improved both quality and speed of K-Means clustering.	Good at handling very large and complex data; Autoencoders catch nonlinear relationships well	Requires heavy computing power and tuning; results are harder to explain
RFM-based segmentation comparing K-Means and Fuzzy C-Means	2021	RFM; K-Means; FCM	The study creates effective customer segments for retention marketing with an easily interpretable performance benchmark showing this new centroid selection can optimize K-Means vs the Fuzzy C-Means algorithm.	Introducing a new custom optimization approach to initial K-Means centroid selection directly addresses the historical challenge of random initialization associated with the K-Means algorithm, yet neatly builds from and leverages the gold-standard RFM (Recency, Frequency, Monetary) business construct.	Evaluation metrics of both methods focus solely on execution speed and mathematical compactness (iterations / runtime) with zero examination of the generated customer segments we would be deploying from a marketing effectiveness or revenue impact standpoint.
Comparative Analysis of the Applicability of Five Clustering Algorithms for Market Segmentation	2023	OPTICS, DBSCAN, Agglomerative, BIRCH, K-Means	The comparative analysis showed that, on this dataset Agglomerative Hierarchical Clustering is superior to the other competitors and illustrated how new implementations of Gowers distance are an optimum when underlying algorithms densitybased or connectivity-based work with hybrid (mixed) datatypes.	It provides a holistic baseline by evaluating three different clustering strategies and utilizes stringent performance metrics —the Davies-Bouldin score and the Silhouette scores— to programmatically confirm segment quality.	Much of the empirical results are strongly reliant on a single open-source hotel industry dataset, indicating that performance conclusion and algorithm superiority ranking probably not directly generalizable to other commercial sectors
An exploration of clustering algorithms for customer segmentation in the UK retail market	2023	RFM; PCA; K-Means; GMM; DBSCAN; Agglomerative Clustering; BIRCH	GMM achieved the best performance with a Silhouette Score of 0.80, outperforming all other clustering algorithms.	Comprehensive comparison of multiple clustering algorithms; PCA improved dimensionality reduction and interpretability; high clustering accuracy using GMM.	Study focused only on one retail dataset; results may not generalize to other industries or real-time systems.
Comparative analysis of machine learning models for customer segmentation	2023	K-Means; DBSCAN; Agglomerative Clustering; PCA + K-Means	Agglomerative Clustering achieved the highest Silhouette Score of 0.6865, producing the best customer segmentation quality.	Compared several machine learning approaches; useful for behavioral-based customer analysis; simple and interpretable evaluation process.	Limited dataset features; evaluation relied mainly on Silhouette Score without additional validation metrics.
Customer Analysis Using ML-Based Algorithms for Effective Segmentation Using RFMT	2023	RFMT + K-Means, GMM, Hierarchical, DBSCAN + Voting	K-Means delivered the most stable clusters. The study successfully identified useful groups like " Champions and At-Risk customers".	Uses real e-commerce data with strong validation; very practical for marketing	Takes a lot of computing power due to voting; struggles with messy data

TABLE II: The Papers Overview

Paper name	Year	Method used	Result summary	Advantage	Disadvantage
Enhancing Pattern Classification Accuracy	2024	APK-Means++ (A smarter version of K-Means)	The study found that this new hybrid model is much more accurate at finding hidden shopping patterns than the basic version.	It is very stable and does a better job of picking the "starting points" for groups so the results are more reliable.	does a better job of picking the "starting points" for groups so the results are more reliable. Because it is a "double" model, it takes a bit more computer time and power to finish the work.
Research on Mall Customers Segmentation Based on K-Means and DBSCAN	2024	K-Means created clearer and more useful customer groups than DBSCAN.	Simple, easy to understand, and produces actionable customer personas	Too small dataset with only two features; oversimplifies real behavior	Limited features
Optimization of Customer Segmentation in Telecommunications (IEEE)	2024	K-Means, DBSCAN, Agglomerative Clustering, Silhouette Analysis	It successfully identified "at-risk" customers who were planning to leave their phone company	It is a very practical study that shows how to turn complex math into a real plan to save a business money.	The data used is very "messy," so you have to spend a lot of time cleaning it before the models will work.
Customer Segmentation Using K-Means Clustering	2024–2025	RFM; K-Means; Mini-Batch; Silhouette	The segment sizes of the five core customers segments, Careless, Careful, Standard Target and Sensible , generates as Results but Mean Shift captures two more high-value behaviors: High Buyers/Frequent Visitors and High Buyers/Occasional Visitors	Compares three different algorithms (K-Means, Agglomerative and Mean Shift) using plausible KPI efforts to uncover latent customer purchasing behavior which is represented multi-dimensionally in terms of annual spending versus visit frequency	Tests a very small dataset of only 200 samples and two features, limiting scalability, real-world complexity and statistical significance to larger businesses
ACM ICIIT (UCI Retail II)	2025	Auto Optimization; K-Means	Best cluster count found	Real-time capable	Depends on scaling
Implementation of Customer Segmentation Model using K-Means and DBSCAN for Fashion Industry Product Transaction	2025	CRISP-DM; StandardScaler; PCA; K-Means; DBSCAN	K-Means outperformed DBSCAN with Silhouette Score = 0.549 and Davies-Bouldin Index = 0.593, while DBSCAN achieved 0.29 and 0.92 respectively.	Applied preprocessing techniques effectively; practical deployment using FastAPI and React dashboard; useful for fashion industry analytics.	Limited comparison to only two clustering algorithms; dataset came from a single company, reducing generalization.
Enhancing Customer Segmentation Through Factor Analysis of Mixed Data (FAMD)-Based Approach Using K-Means and Hierarchical Clustering Algorithms	2025	FAMD + K-Means + Hierarchical Clustering	The method handled both numerical and categorical data well, resulting in more stable and meaningful customer segments.	Excellent for mixed-type customer data (similar to your datasets)	Adds extra computation step; sensitive to parameter selection
Bank Segmentation via Improved DBSCAN	2025	KM-DBSCAN (Banking specific)	This method helped a bank increase its money by 16% because it targeted the exact right customers for loans.	It uses a special "automatic" logic that calculates the best settings for you, so you don't have to guess or use trial and error.	The way this model is set up is mostly for banks, so it might need to be changed to work for other businesses.

III. PROPOSED METHODOLOGY

The methodology proposed in this study adopts a systematic unsupervised machine learning framework for customer segmentation and behavioral analysis using comprehensive e-commerce and retail datasets. The process begins with data acquisition and integration from multiple sources, followed by thorough data preprocessing steps including handling missing values, outlier detection, data cleaning, and encoding of categorical variables. Exploratory Data Analysis (EDA) is then conducted to uncover initial patterns, distributions, and correlations among key attributes such as demographics, purchase behavior, online engagement, and loyalty metrics. Feature engineering and selection are performed to create meaningful variables (e.g., RFM scores, engagement indices, and spending tiers) and reduce dimensionality using techniques such as PCA when necessary. Feature scaling and normalization are applied to ensure compatibility with distance-based algorithms. For segmentation, multiple clustering algorithms including K-means, Hierarchical Clustering, and Gaussian Mixture Models are implemented and compared, with the optimal number of clusters determined through elbow method, silhouette analysis, and Davies-Bouldin index. Finally, the resulting customer segments are profiled and interpreted using descriptive statistics and visualization techniques to derive actionable business insights, followed by validation against domain knowledge.

A. Datasets Descriptions

A dataset description serves as the foundational context and comprehensive guide for every dataset in our collection. It provides essential information about what the data represents, how it was collected, its scope, and its potential applications.

A. Online Retail Customer Dataset The Online Retail Customer Dataset contains 9,000 customer records with rich behavioral attributes, making it highly suitable for customer segmentation and unsupervised machine learning in the e-commerce domain. It includes key information on customer demographics (age, gender, income, region), purchasing behavior (order frequency, total spend, product categories, return rates), loyalty metrics, and online engagement (session duration, page views, cart abandonment). This dataset enables the identification of distinct customer segments such as high-value buyers, price-sensitive shoppers, and at-risk customers. It supports comprehensive analysis workflows including RFM modeling, feature engineering, clustering (K-means, Hierarchical, DBSCAN, GMM), and dimensionality reduction, ultimately helping businesses design targeted marketing, personalized recommendations, and customer retention strategies.

TABLE III: Features of Online Retail Customer Dataset

Features	Type	Description
age	Numerical	Customer age
Annual_Income_USD	Numerical	Annual customer income
spending _frequency	Numerical	Numbering of purchases made
spending _score	Numerical	Customer spending behavior score
Membership _Status	Categorical	Customer loyalty membership level
Region	Categorical	Customer geographical region
Total_Purchases	Numerical	Total number of purchases made
Website_Visits _Last_Month	Numerical	Number of visits to the website last month
Avg_Time_Per _Visit_Minutes	Numerical	Average time spent on website per visit
Referred _Friends	Numerical	Number of friends referred to the platform

B. Customer Purchase Behavior (E-Commerce)

The Customer Purchase Behavior (E-Commerce) dataset is a large-scale and highly valuable resource consisting of nearly 500,000 records collected directly from an e-commerce platform, offering deep visibility into how customers interact with online shopping services. It encompasses a rich array of features including demographic details (age, gender, location, income), browsing behavior (session duration, pages viewed, time spent per category, device type), purchase history (order frequency, average order value, total spend, product categories, purchase timing), engagement signals (cart abandonment, wishlist activity, click-through rates), and marketing analytics attributes (campaign exposure, discount usage, response to promotions). This massive volume of data makes it exceptionally powerful for machine learning applications, particularly customer segmentation, clustering analysis, behavioral pattern discovery, and Customer Lifetime Value (CLV) prediction. Analysts can uncover distinct customer groups such as high-value frequent buyers, bargain hunters, window shoppers, brand-loyal enthusiasts, or at-risk churners.

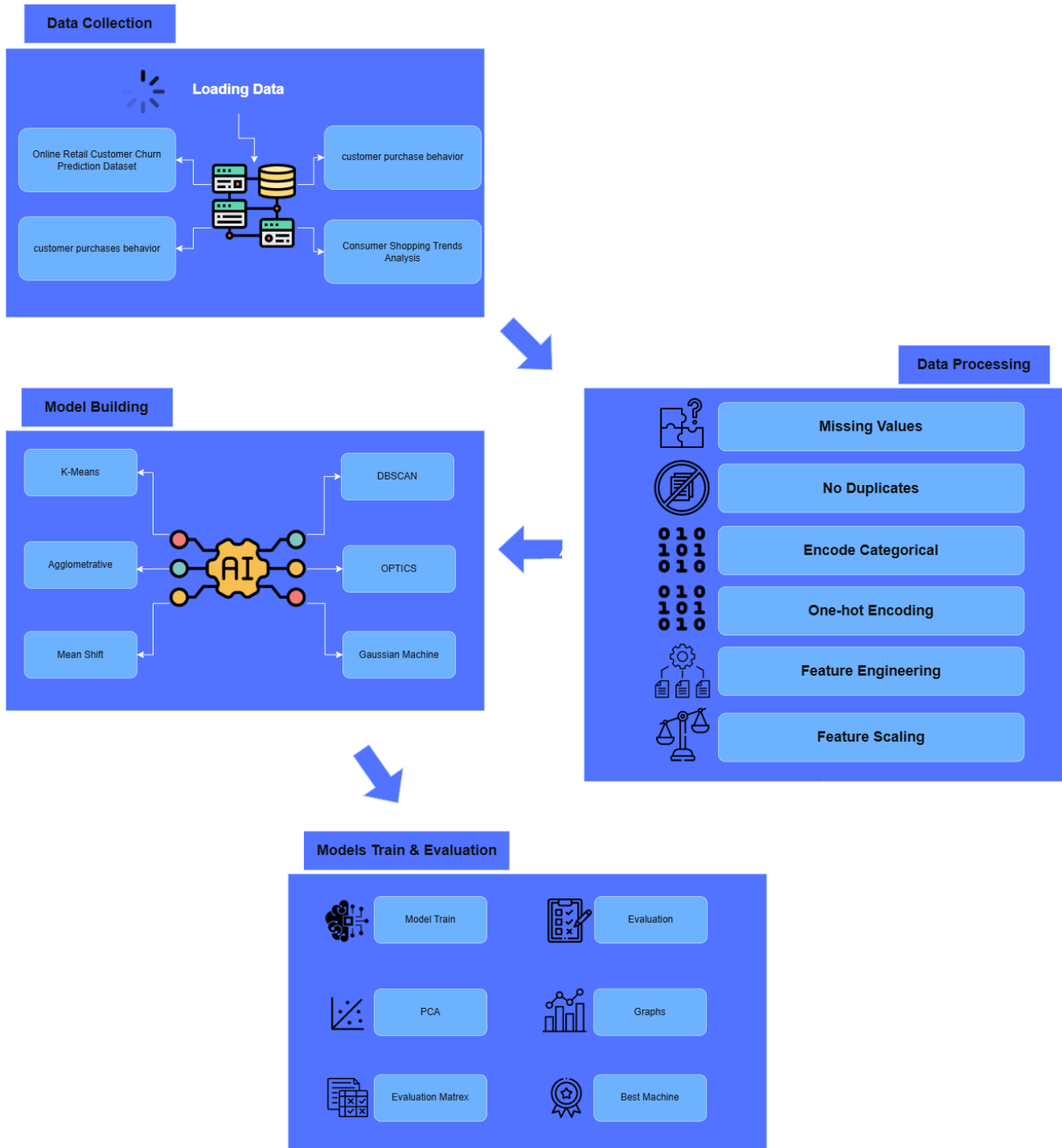


Fig. 1: Overall Proposed Methodology for Customer Segmentation and Clustering Analysis

TABLE IV: Detailed Features of the Customer Purchase Behavior Dataset

Feature	Type	Description
age	Integer	Customer's age in years. Range: 15–81.
income	Float	Customer's annual income in USD. Range: 11,966 – 204,178.
purchase_frequency	Integer	Total purchases made by the customer. Indicates engagement level.
time_on_website	Float	Average time (in minutes) spent on the website per visit.
tenure_years	Float	Duration (in years) since the customer joined the platform.
loyalty_program	Binary	Whether the customer is enrolled in a loyalty program (0/1).
discounts_availed	Integer	Number of discounts/coupons redeemed by the customer.
session_count	Integer	Number of sessions/visits recorded for the customer.
customer_satisfaction	Integer	Satisfaction rating scale (1–5).
spending_score	Float	Calculated metric representing customer spending behavior.
loyalty_score	Float	Calculated metric representing customer brand loyalty.
online_activity	Float	Composite metric of customer engagement on the platform.
region	Categorical	Geographic region of the customer (North, South, East, West).

C. Consumer Shopping Trends Analysis dataset

The Consumer Shopping Trends Analysis dataset stands as a robust and versatile resource comprising 11,789 customer records with 25 detailed behavioral and demographic attributes, offering deep insights into modern retail dynamics by blending demographic information (age, income, gender, region), transactional behaviors (purchase frequency, spending score, average order value), loyalty indicators (loyalty score, subscription status), digital engagement (online activity level, browsing time, cart abandonment rate), and omnichannel preferences (online vs in-store purchases, preferred payment methods, and shipping choices). This rich combination enables advanced customer segmentation and clustering analysis—such as identifying high-value loyalists, price-sensitive young online shoppers, frequent in-store regional buyers, or emerging digital natives—through algorithms like K-means, hierarchical clustering, or Gaussian Mixture Models, while supporting extensive machine learning workflows including data preprocessing, feature scaling, outlier handling, dimensionality reduction with PCA, and predictive modeling for churn, lifetime value, or next-purchase prediction. Engineered from seven core features (age, income, purchase frequency, loyalty score, region, spending score, and online activity), the dataset mirrors real-world consumer environments with both numerical and categorical variables, making it ideal for uncovering hidden behavioral patterns, regional spending differences, and the growing impact of digital channels on traditional retail.

TABLE V: Features of Consumer Shopping Trends Dataset

Feature	Type	Description
age	Numerical	Customer age
income	Numerical	Monthly customer income
purchase_frequency	Numerical	Number of monthly online orders
loyalty_score	Numerical	Customer brand loyalty level
region	Categorical	Customer city tier classification (Tier 1/2/3)
gender	Categorical	Customer gender
spending_score	Numerical	quantifies the overall purchasing power and spending behavior
online_activity	Numerical	measures the overall digital engagement level of each customer
avg_online_spend	Numerical	amount of money customer spends online
avg_store_spend	Numerical	amount of money customer spends physically
discount_sensitivity	Numerical	degree to which customer is influenced by discounts and offers during purchasing.
daily_internet_hours	Numerical	avg hours customer spends online per day
social_media_hours	Numerical	avg time customer spends on social media
tech_savvy_score	Numerical	level of digital literacy and technological proficiency of customer
online_payment_trust_score	Numerical	level of trust customer has in online payment systems and digital transactions

D. Customer Purchases Behaviour Dataset

The Customer Purchase Behaviour Dataset is a rich and comprehensive collection comprising 100,001 customer records, offering deep insights into actual shopping behaviors within a modern retail and e-commerce environment. Unlike traditional datasets that focus primarily on transactional logs, this dataset captures a wide spectrum of nuanced behavioral signals, including visit frequency, session duration, return patterns, geographic location, loyalty program participation, promotional responsiveness, browsing habits, cart abandonment rates, and detailed spending patterns. By integrating both demographic attributes and dynamic behavioral indicators, it provides a holistic view of customer intent, preferences, and decision-making processes. This extensive dataset is particularly valuable for unsupervised machine learning applications, especially customer segmentation and behavioral profiling. It enables analysts to uncover hidden customer personas and naturally occurring segments by identifying groups that share similar motivations, habits, and purchasing journeys. Through advanced clustering algorithms such as K-Means, Hierarchical Clustering, Gaussian Mixture Models (GMM), or DBSCAN, distinct customer types can be effectively differentiated — for

instance, high-income professionals making infrequent but high-value purchases, budget-conscious students who actively chase promotions, loyal local residents with frequent repeat visits, price-sensitive occasional shoppers, and highly engaged brand advocates. Furthermore, the dataset supports sophisticated analyses including RFM modeling (Recency, Frequency, Monetary), Customer Lifetime Value (CLV) prediction, churn risk assessment, and personalized marketing strategy development. Its balanced mix of numerical, categorical, and behavioral features makes it an excellent resource for feature engineering, dimensionality reduction, and deriving actionable business intelligence in competitive retail landscapes.

Features	Type	Description
age	Numerical	Represents the customer's life stage and potential needs.
income	Numerical	The financial capacity and potential purchasing power of the shopper.
purchase_amount	Numerical	The total monetary value the customer has invested in the brand.
purchase_frequency	Categorical	A measure of habit, labeling how often the customer thinks of the brand.
region	Categorical	Provides the geographic context and regional preferences of the user.
satisfaction_score	Numerical	The rating of their experience.
promotion_usage	Numerical	An indicator of price sensitivity and engagement with marketing deals.
loyalty_status	Categorical	The formal level of commitment the customer has reached (e.g., Gold, Silver).

B. Data Preprocessing

For customer segmentation and clustering analysis, preprocessing is a very important step in which the raw data of customer behavior is transformed into a structured and numerical form. Also, it is required to make Machine Learning clustering algorithms to use the dataset efficiently. A number of measures are undertaken to improve the dataset quality and to prepare it for clustering analysis and customer behaviour profiling.

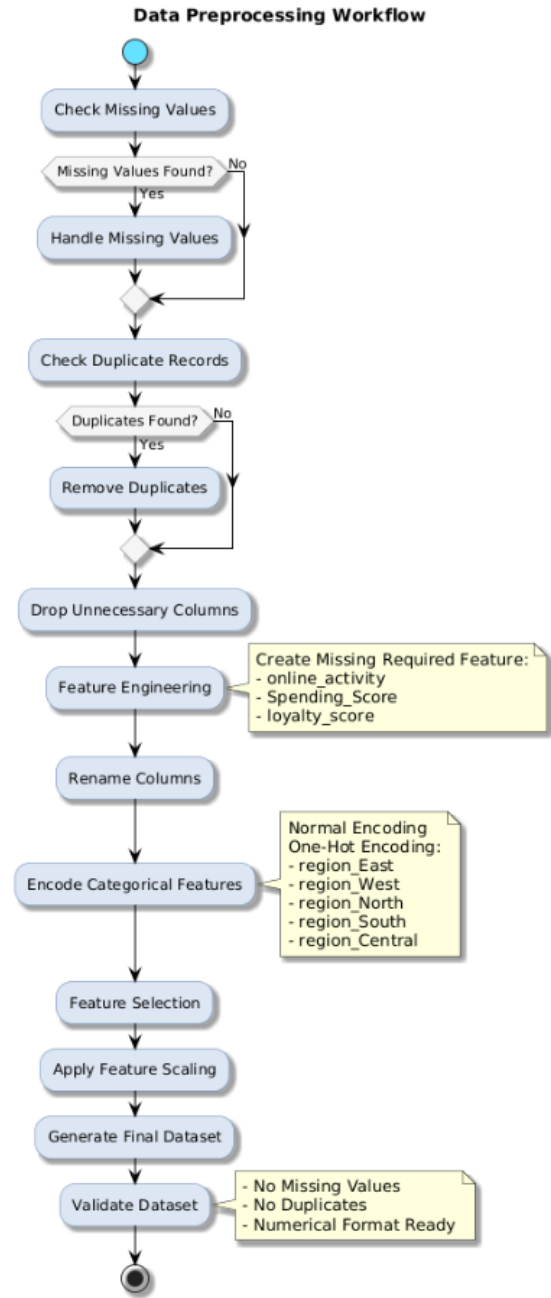


Fig. 2: Preprocessing Steps

1) Datasets: Preprocessing Steps:

1) Data Loading:

The first dataset: Online Retail Customer Dataset
 The first step involved downloading the Online Retail Customer dataset from Kaggle. The dataset was structured as a CSV file with 9,000 customer records. The dataset includes a variety of numerical and categorical features concerning customer demographics, purchasing habits, loyalty tiers, and online engagement actions. The dataset was imported using the pandas library for further preprocessing and clustering analysis .

The second dataset: Customer Purchase Behavior Dataset (E-Commerce) was loaded from a CSV file containing 500,000 customer records. Due to computational constraints, a random sample of 10,000 records was extracted using a fixed random state to ensure reproducibility. The dataset contains numerical and categorical features related to customer demographics, purchasing behavior, loyalty levels, and regional distribution. The dataset was loaded using the pandas library and prepared for preprocessing and clustering analysis.

The Third dataset: Consumer Shopping Trends Dataset step includes loading the dataset from Kaggle, which was formatted as a CSV file containing 11,789 customer records. The dataset contains multiple numerical and categorical features related to customer demographics, purchasing behavior, loyalty levels, and digital engagement activities. The dataset was loaded using the pandas library for further preprocessing and clustering analysis.

The Fourth dataset: Customer Purchases Behaviour Dataset. It contains 100,001 customer records, where each row represents a customer and their purchasing behavior within the retail environment.

The dataset combines numerical and categorical features that reflect different aspects of customer behavior, including age, income, purchase patterns, loyalty level, and engagement with promotions. This makes it useful for understanding how customers differ from one another, not only in spending but also in how frequently they interact with the business.

The datasets was loaded from a CSV file using pandas, allowing the data to be organized into a dataframe and prepared for preprocessing, feature engineering, and clustering analysis.

2) Missing Values Handling:

To ensure data completeness before applying clustering algorithms, the dataset was tested for missing or null value. Missing values may negatively affect the quality of clustering and distance calculations [8]. Therefore, all dataset columns were validated and confirmed to have missing values, the dataset is suitable for preprocessing and machine learning operations.

3) Duplicate Records Removal:

To prevent redundancy and biased clustering results, duplicate customers were checked and removed. Removing duplicate rows helps to improve the clustering accuracy [6] and ensure that each customer contributes equally to the clustering process without

repeated information.

4) Date Column Conversion:

Columns related to data such as last purchase date were processed and converted into a suitable format to facilitate preprocessing operations. Data conversion enables machine learning algorithms to handle temporal information efficiently and prevents incompatible data types for clustering analysis.

5) Feature Engineering:

The first dataset: A new feature **online_activity** was generated to represent customer digital engagement behavior. This feature was created by aggregating website visits, average session duration, and referral activities into a single engagement score. Feature engineering improves the quality of clustering by extracting more meaningful behavioral indicators from the raw dataset [5] [17]

The formula for the **online_activity** is $(0.50 \times \text{Website_Visits_Last_Month}) + (0.35 \times \text{Avg_Time_Per_Visit_Minutes}) + (0.15 \times \text{Referred_Friends})$

The Second Dataset :

The Customer Purchase Behavior Dataset (E-Commerce) dataset was imported from a CSV file containing 500,000 customer records. We extracted a random sample of 10,000 records using a fixed random state for reproducibility. The data was imported using the pandas library and prepared for clustering analysis.

$$\text{Spending Score} = \left(\frac{\text{income}}{\max(\text{income})} \times 50 \right) + \left(\frac{\text{purchase_frequency}}{\max(\text{purchase_frequency})} \times 30 \right) + \left(\frac{\text{discounts_availed}}{\max(\text{discounts_availed})} \times 30 \right)$$

$$\text{Loyalty Score} = \left(\frac{\text{loyalty_program}}{\max(\text{loyalty_program})} \times 40 \right) + \left(\frac{\text{customer_satisfaction}}{\max(\text{customer_satisfaction})} \times 40 \right) + \left(\frac{\text{tenure_years}}{\max(\text{tenure_years})} \times 20 \right)$$

$$\text{Online Activity} = \left(\frac{\text{session_count}}{\max(\text{session_count})} \times 40 \right) + \left(\frac{\text{time_on_website}}{\max(\text{time_on_website})} \times 40 \right) + \left(\frac{\text{purchase_frequency}}{\max(\text{purchase_frequency})} \times 20 \right)$$

Third dataset: Two new features for the clustering process were engineered.[14, 15].

The first engineered feature, **spending_score**, was created to represent the overall spending behavior of each customer.[15]

Spending Score

$$\begin{aligned} \text{spending_score} = & \left(\frac{\text{income}}{\max(\text{income})} \times 35 \right) \\ & + \left(\frac{\text{avg_online_spend} + \text{avg_store_spend}}{\max(\text{avg_online_spend} + \text{avg_store_spend})} \times 35 \right) \\ & + \left(\frac{\text{purchase_frequency}}{\max(\text{purchase_frequency})} \times 20 \right) \\ & + \left(\frac{\text{discount_sensitivity}}{\max(\text{discount_sensitivity})} \times 10 \right) \end{aligned}$$

The second engineered feature, **online_activity**, was designed to capture the digital engagement behavior of each customer.[14]

Online Activity

$$\begin{aligned} \text{online_activity} = & \left(\frac{\text{daily_internet_hours}}{\max(\text{daily_internet_hours})} \times 35 \right) \\ & + \left(\frac{\text{social_media_hours}}{\max(\text{social_media_hours})} \times 30 \right) \\ & + \left(\frac{\text{tech_savvy_score}}{\max(\text{tech_savvy_score})} \times 20 \right) \\ & + \left(\frac{\text{online_payment_trust_score}}{\max(\text{online_payment_trust_score})} \times 15 \right) \end{aligned}$$

The Fourth data set: Several derived features were created from the original Customer Purchases Behaviour Dataset in order to improve the quality of customer segmentation. These engineered variables were designed to capture behavioral patterns better that were not explicitly available in the raw dataset.

Spending Score

$$\begin{aligned} \text{Spending Score} = & \left(\frac{\text{Income}}{\max(\text{Income})} \times 40 \right) \\ & + \left(\frac{\text{Purchase Amount}}{\max(\text{Purchase Amount})} \times 40 \right) \\ & + \left(\frac{\text{Purchase Frequency}}{3} \times 20 \right) \end{aligned}$$

Online Activity

$$\begin{aligned} \text{Online Activity} = & \left(\frac{\text{Promotion Usage}}{\max(\text{Promotion Usage})} \times 40 \right) \\ & + \left(\frac{\text{Purchase Frequency}}{3} \times 30 \right) \\ & + \left(\frac{\text{Satisfaction Score}}{\max(\text{Satisfaction Score})} \times 30 \right) \end{aligned}$$

6) Feature Normalization:

The features used in the online activity score calculation

are normalized before aggregation to ensure fair contribution of each feature. Normalization prevents features with larger numerical ranges to dominate the clustering process, and improves the performance of distance-based clustering algorithms such as K-Means and DBSCAN [18].

7) Column Renaming:

In the dataset preprocessing, several columns were renamed to cleaner and more readable formats to improve the consistency and clarity. For example, we can convert `Annual_Income_USD` to `income` and `Spending_Score` to `spending_score`. This makes the text more readable and makes it easier to handle the feature in the clustering analysis.

8) Categorical Feature Encoding:

Categorical variables like customer regions were transformed to numerical variables using One-Hot Encoding. In this preprocessing step, separate binary columns were created for each region category (East, West, North, South, Central) [9]. Clustering algorithms need numeric data as input, so we have to encode the categorical variables.

9) Feature Scaling:

All numerical features were scaled using the **StandardScaler** technique for feature scaling. Data set scaling made data in common range with zero mean and one variance which improved the quality of clustering and gave the fair distance measurement for all the features

10) Final Dataset Validation:

Post performing the preprocessing operations, the dataset was validated for the readiness of clustering algorithms. The final dataset had no missing values, no duplicate records and all features were converted into numerical formats [10] suitable for unsupervised machine learning and clustering analysis.

C. Data Visualization

Using visualization makes working with customer behavior data much easier, especially in areas like customer segmentation and clustering analysis. Transforming complex customer patterns into visually understandable graphs and charts makes it easier to identify hidden behaviors, purchasing trends and customer engagement patterns. Without visual representations it would be difficult to understand relationships between income, spending behavior, loyalty levels and online activity in large-scale retail datasets. Visualization techniques aid understanding of the data, help in making better business decisions and in assessing the quality of the clustering process[3]. This allows organizations to better understand customer groups,

improve marketing strategies and enhance customer targeting and retention .

1) Dataset 1 Visualization:

1) Customer Age Distribution:

The distribution of customers by age was visualized to analyze the variation of the ages of customers in the dataset. The frequency of customers in various age ranges was presented in a histogram.

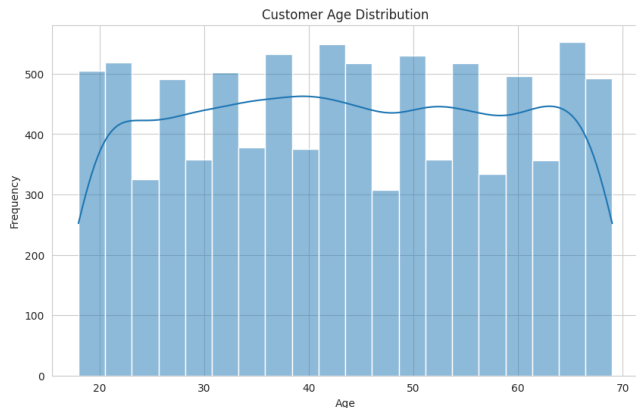


Fig. 3: Customer Age Distribution

2) Annual Income Distribution:

To see the distribution of financials amongst customers we visualized the annual income feature using a histogram. This graph shows the distribution of customer income and helps to find low income and high income groups of customers. Knowledge of income distribution is necessary for customer segmentation and analysis of purchasing behaviour.

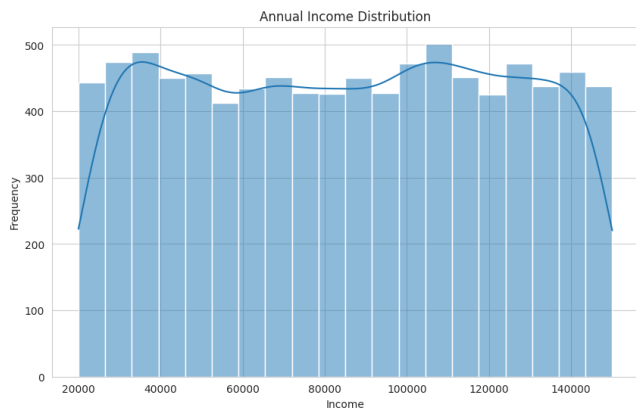


Fig. 4: Annual Income Distribution

3) Spending Score Distribution:

The distribution of spending scores was visualized to analyze the customer purchasing behavior patterns. Spending score indicates the level of active spending of customers within the retail platform. The visualization

helps to identify customers with high spending behavior and customers with limited purchasing activity.

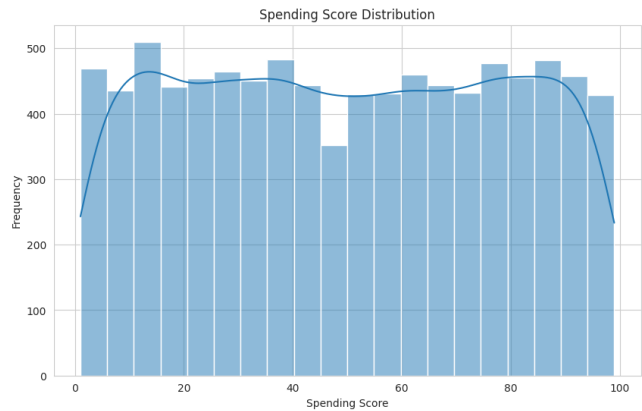


Fig. 5: Spending Score Distribution

4) **Region Distribution:** For this, a count plot was plotted to analyze the distribution of customer distribution across the geographical regions. The graph shows the concentration of customers in areas such as East , West , North , South and Central . This visualization helps in regional customer analysis and business expansion planning.

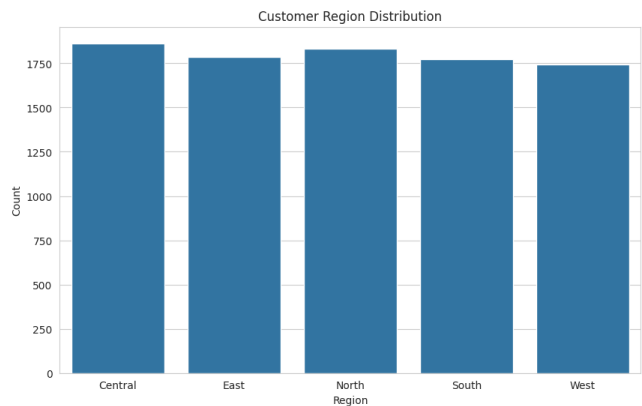


Fig. 6: Customer Region Distribution

5) Online Activity Distribution:

The online activity feature was visualized for analyzing customer digital engagement behavior. The graph shows the level of customer interaction based on website visits, average session length and referral activities. This visualization can be used to identify highly engaged customers and inactive customer segments.

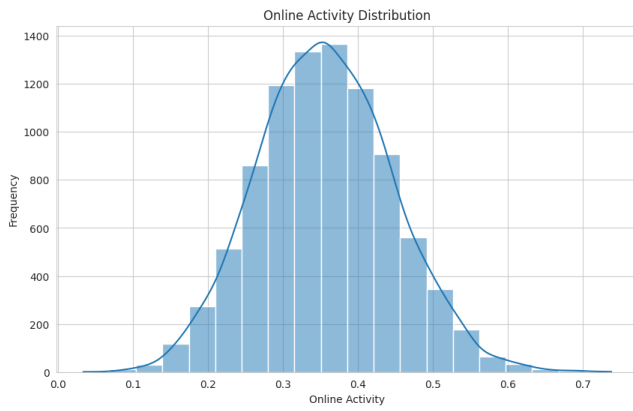


Fig. 7: Online Activity Distribution

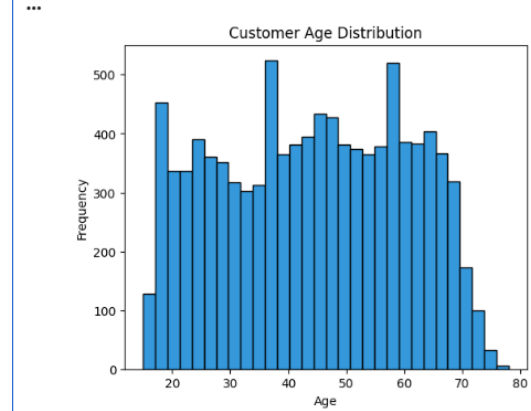


Fig. 9: Customer Age Distribution.

6) Feature Correlation Heatmap:

Correlation heatmap was generated to understand the relationship between numerical features in the dataset. The heatmap shows the positive and negative correlation between variables such as income, spending score, purchase frequency and online activity. This visualization helps to understand the dependencies between features and supports clustering analysis .

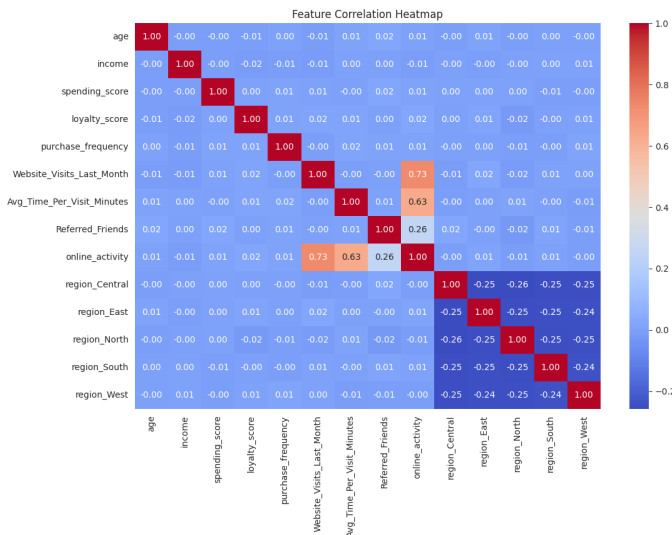


Fig. 8: Feature Correlation Heatmap

2) *Dataset 2 Visualization:* To gain a comprehensive understanding of the customer base and behavioral patterns, several visualizations were performed on the dataset features.

1) **Customer Age Distribution:** The distribution of customers by age was visualized to analyze the variation of customer ages within the dataset. A histogram was used to display the frequency of customers across different age ranges, revealing a broad engagement across ages 15–81, with notable peaks in young adulthood and the mid-50s.

2) **Annual Income Distribution:** The annual income feature was visualized using a histogram to examine the financial

distribution of customers. The graph illustrates the spread of customer income levels, showing a steady customer base in middle-income brackets with a decline in higher earners above \$150,000. Knowledge of income distribution is necessary for customer segmentation and purchasing behaviour analysis.

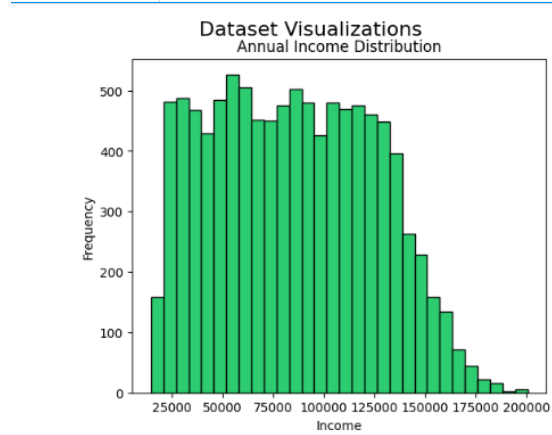


Fig. 10: Annual Income Distribution.

3) **Spending Score Distribution:** The spending score feature was visualized using a histogram to understand the overall spending tendencies of customers. The distribution follows a near-normal pattern, identifying a central tendency for moderate spending behaviour among the majority of customers.

4) **Purchase Frequency Distribution:** The purchase frequency feature was visualized to examine how often customers engage in shopping activity. The resulting distribution shows distinct clusters of shopping habits, clearly distinguishing between occasional and frequent purchasers within the dataset.

5) **Loyalty Score Distribution:** The loyalty score feature was visualized to assess the degree of customer retention and engagement. The bimodal distribution observed clearly separates low-loyalty casual users from high-loyalty power users, reflecting two distinct behavioural segments within the customer base.

6) **Customer Region Distribution:** The Customer Region Distribution analysis reveals the geographical spread of the

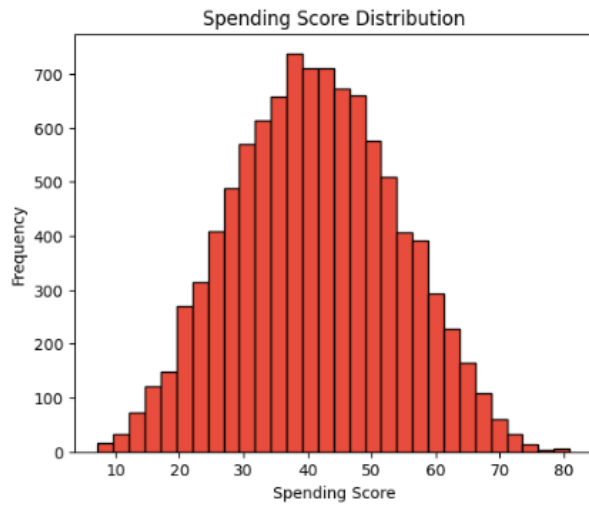


Fig. 11: Spending Score Distribution.

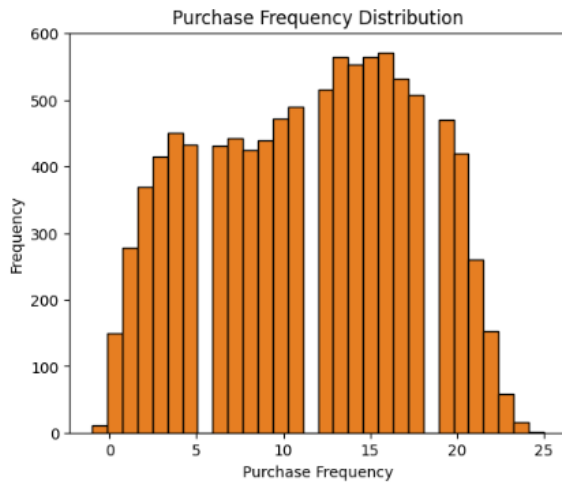


Fig. 12: Purchase Frequency Distribution.

customer base across different regions, providing critical insights into market concentration and regional performance on the platform. The visualization clearly highlights the **South** as the most dominant region, accounting for the largest share of customers, followed by other regions such as the North, West, and East in descending order. This strong southern concentration suggests several possible factors, including higher population density, greater e-commerce adoption, stronger marketing presence, favorable economic conditions, or better logistics and delivery infrastructure in southern areas. Understanding this distribution is essential for customer segmentation, as it allows businesses to identify region-specific behavioral patterns — for instance, whether southern customers show higher spending scores, different product preferences, or greater loyalty compared to customers from other regions. These geographic insights support more effective decision-making in areas such as targeted regional marketing campaigns, localized promotions, inventory allo-

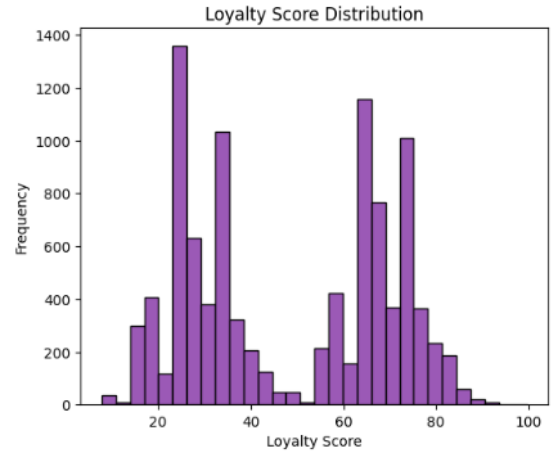


Fig. 13: Loyalty Score Distribution.

cation, expansion strategies, and resource optimization, ultimately helping retailers maximize reach in high-potential areas while addressing underpenetrated regions for balanced national growth.

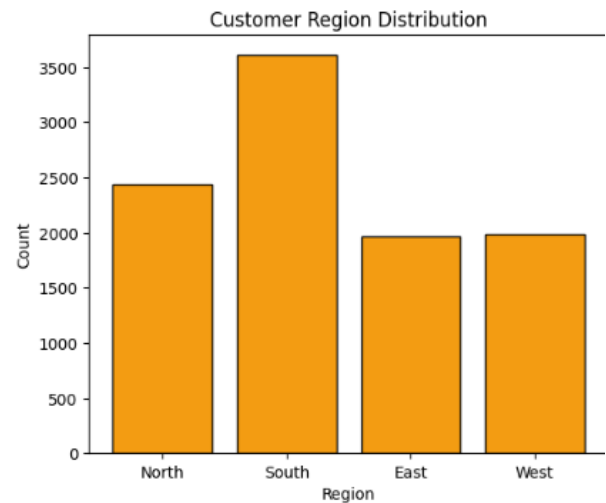


Fig. 14: Customer Region Distribution.

7) Online Activity Distribution: The engineered online activity feature was visualized to examine digital engagement behaviour across customers. The distribution analyzes session frequency and interaction levels, offering insight into how actively customers engage with the platform online.

8) Feature Correlation Heatmap: A correlation heatmap was generated to examine the statistical relationships between all numerical features in the dataset. The heatmap statistically validates a strong positive relationship of 0.77 between customer income and spending scores.

3) Dataset 3 Visualization:

1) **Customer Age Distribution:** The age distribution of customers was visualized to analyze the variation of customer ages within the dataset. A histogram was used to display the frequency of customers across different

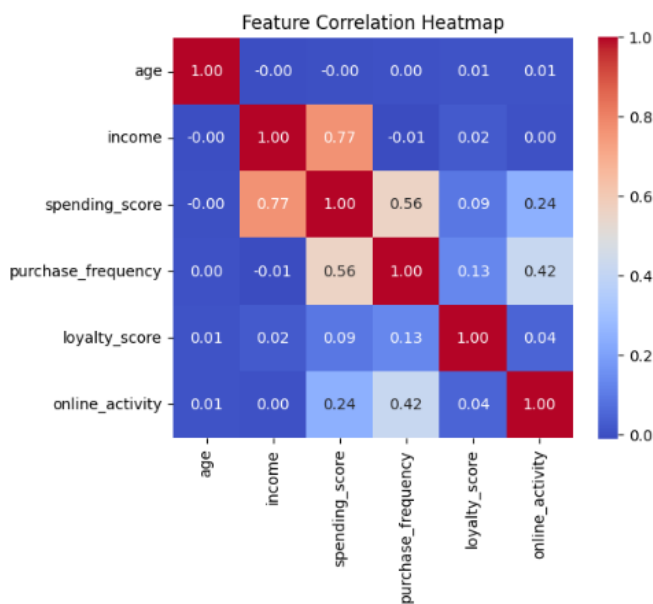


Fig. 15: Feature Correlation Heatmap.

age ranges. This visualization helps identify dominant customer age groups and provides insights into the demographic structure of the consumer shopping environment.

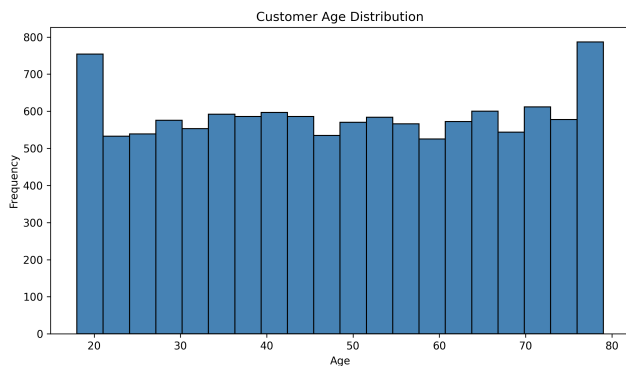


Fig. 16: Customer Age Distribution

2) **Monthly Income Distribution:** The monthly income feature was visualized using a histogram to examine the financial distribution of customers. This graph illustrates the spread of customer income levels and helps identify low-income and high-income customer groups within the dataset.

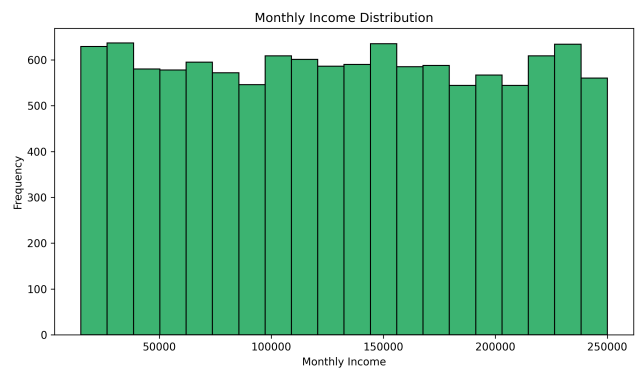


Fig. 17: Monthly Income Distribution

3) **Spending Score Distribution:** **Spending Score Distribution** provides a clear visualization of customers' overall purchasing power and engagement levels on the platform. This composite feature is intelligently derived from multiple key factors including monthly income, average spending on online and in-store purchases, purchase frequency, and discount sensitivity, offering a holistic measure of customer value. The distribution chart typically reveals distinct groups: high-spending customers who contribute significantly to revenue, moderate spenders who represent the core base, and low-spending customers with limited purchasing activity. This analysis is highly valuable for segmentation as it helps retailers differentiate between premium, high-value buyers and more price-sensitive or occasional shoppers, enabling better targeting of personalized offers, loyalty programs, premium product recommendations, and promotional strategies tailored to each spending tier.

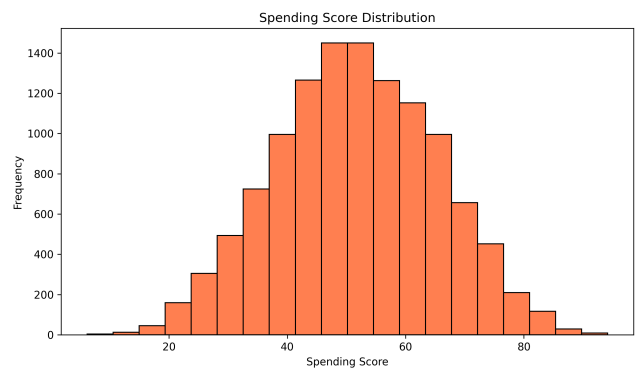


Fig. 18: Spending Score Distribution

4) **City Tier Distribution:** Customer city tier distribution was visualized using a count plot to examine how customers are distributed across urban classification levels. The graph highlights customer concentration within Tier 1, Tier 2, and Tier 3 city categories, representing metropolitan, urban, and semi-urban areas respectively.

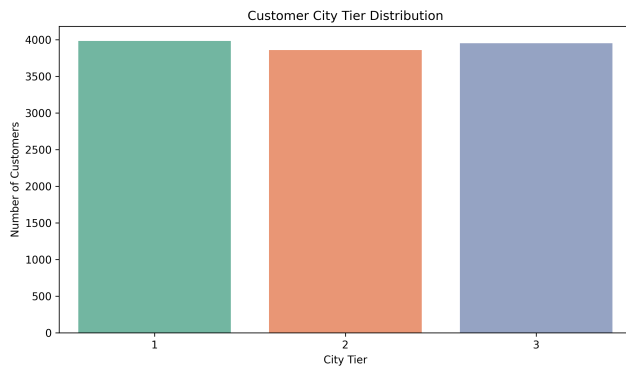


Fig. 19: Customer City Tier Distribution

5) **Online Activity Distribution:** **Online Activity Distribution** offers valuable insights into customers' digital engagement levels on the platform. This composite feature is derived from several key indicators, including daily internet hours, social media usage, tech-savviness score, and online payment trust level, creating a comprehensive measure of each customer's digital behavior. The visualization effectively distinguishes between highly engaged digital customers — who spend significant time online, show high tech confidence, and actively use digital payment methods — and those with lower online interaction levels who prefer traditional shopping channels or have limited digital comfort. This distribution analysis is crucial for understanding the shift toward online shopping, identifying digitally native customer segments, and spotting opportunities to improve digital adoption among less active users. Businesses can leverage these insights for targeted digital marketing campaigns, personalized website experiences, mobile app promotions, and strategies to convert low-engagement customers into more active online shoppers.

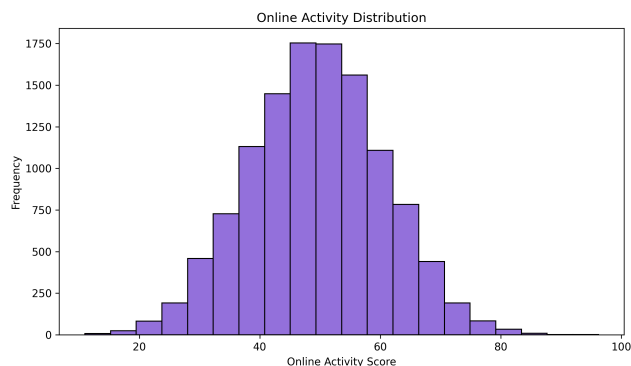


Fig. 20: Online Activity Distribution

6) Feature Correlation Heatmap:

A correlation heatmap was generated to analyze the relationships between the numerical features selected for clustering. The heatmap visually represents positive and negative correlations between variables such as age,

income, spending score, purchase frequency, loyalty score, and online activity.

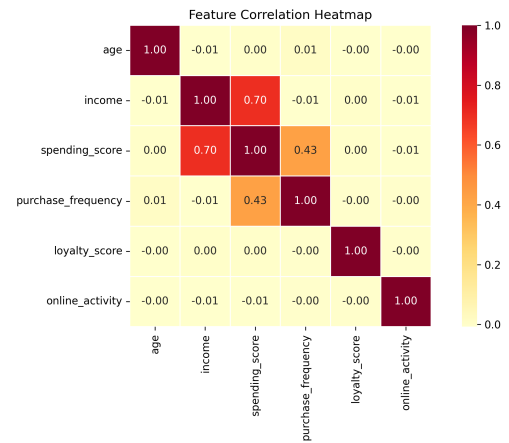


Fig. 21: Feature Correlation Heatmap

4) Dataset 4 Visualization:

1) Age Distribution:

The age distribution graph shows how customer ages are distributed in the dataset. This visualization helps to identify the dominant age groups and gives an overview of the customer age group structure.

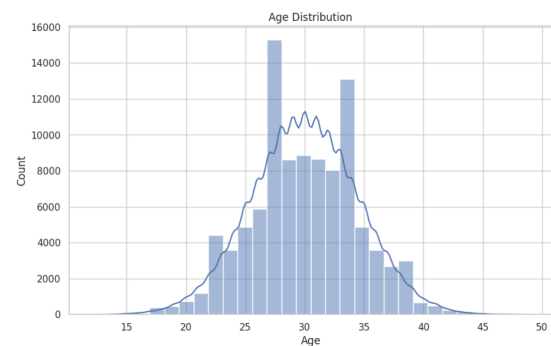


Fig. 22: Age distribution of customers

2) Income Distribution:

The income distribution displays the annual income of the customers in the dataset. It provides an understanding of the buying power of customers and helps in segmenting based on financial ability.

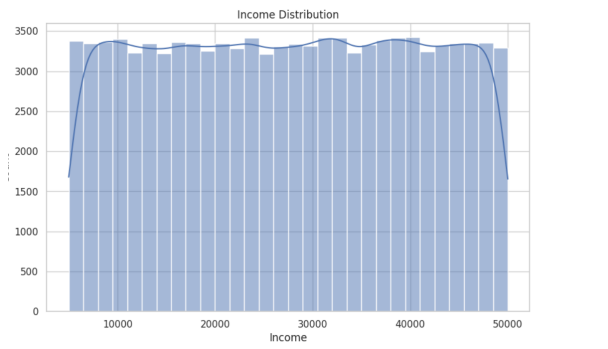


Fig. 23: Income distribution of customers

3) Purchase Amount Distribution:

This graph shows the amount customers spent. It helps to identify spending behavior. It shows difference between low and high purchase customers.

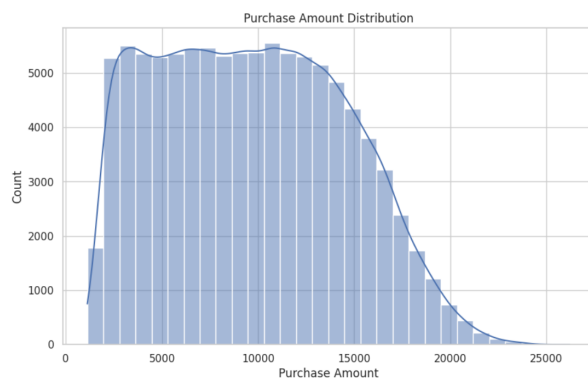


Fig. 24: Purchase amount distribution

4) Region Distribution:

The distribution by region shows the number of customers in different geographical regions. This helps to understand the number of customers in different locations.



Fig. 25: Customer distribution by region

5) Loyalty Status Distribution:

This visualization shows how many customers fall into each loyalty status group. It gives a clear picture of how well we are keeping customers and highlights trends in their loyalty over time.

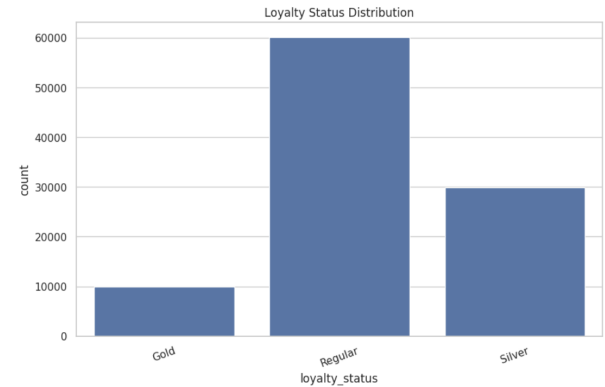


Fig. 26: Loyalty status distribution

6) Purchase Frequency Distribution:

The purchase frequency chart shows how often customers make purchases. This supports understanding of customer engagement and buying behavior.



Fig. 27: Purchase frequency distribution

7) Spending Score Distribution:

This chart shows how customers score when it comes to spending. The score brings together details like how much they earn, how often they buy, and how much they spend, making it easier to understand different shopping habits.

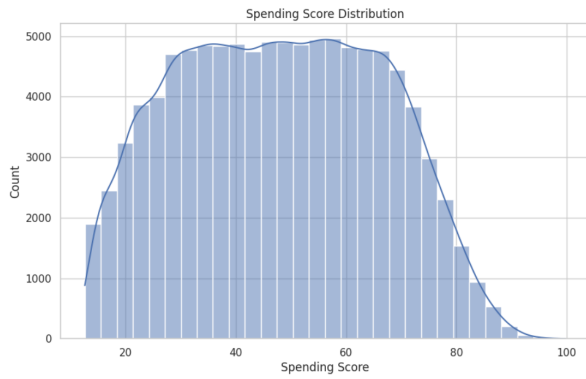


Fig. 28: Spending score distribution

8) Online Activity Distribution:

This graph illustrates the distribution of customers' overall online activity levels, providing a clear view of digital engagement across the dataset. The score is calculated based on key behavioral indicators such as frequency of promotion usage, shopping frequency, and customer satisfaction levels, offering a well-rounded measure of how actively customers engage with the platform in the digital space. The visualization helps distinguish between highly active online customers — who frequently respond to promotions, shop often, and report high satisfaction — and those with lower digital engagement who interact less frequently. These insights are valuable for identifying digitally active segments that respond well to online campaigns while highlighting opportunities to boost engagement among less active customers through improved user experience, targeted promotions, and personalized digital incentives.

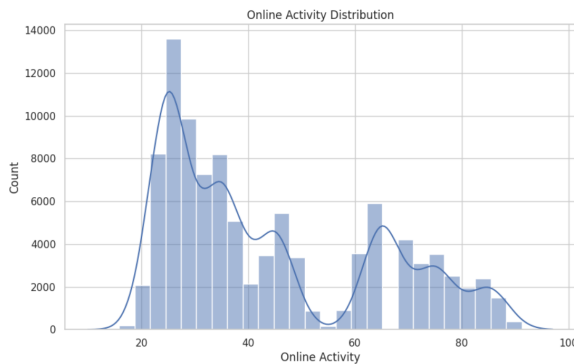


Fig. 29: Online activity distribution

9) Correlation Heatmap:

The correlation heatmap shows relationships among numerical features in the dataset. It identifies positive and negative correlations between customer attributes and validates feature dependencies before clustering.

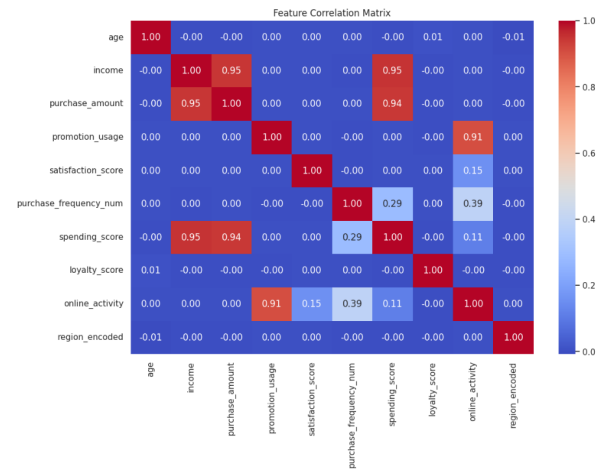


Fig. 30: Correlation heatmap of selected features

D. Clustering Algorithms Visualization

Visualization is very important in clustering analysis because it helps us to understand how clustering algorithms group data into separate groups[18]. Graphical representation of clustered data can be used by researchers to evaluate the quality of clusters, overlaps between clusters and analyze customer segmentation behavior. It also simplifies comparison of the performance of multiple clustering algorithms and understanding their ability to discover hidden patterns inside datasets.

1) Clustering Algorithms Visualization:

1) K-Means Clustering Visualization:

The K-Means clustering visualization illustrates how the algorithm groups data into a specified number of clusters, using the distance from the centroid. Each cluster is represented with different colors for clear visualization. The visualization can be used to evaluate the quality of cluster compactness and separation.



Fig. 31: K-Means Clustering Visualization

2) Agglomerative Clustering Visualization:

Agglomerative clustering visualization illustrates hierarchical grouping of data points according to similarity measures. The visualization shows cluster formation via merging operations, which helps to understand hierarchical relationships

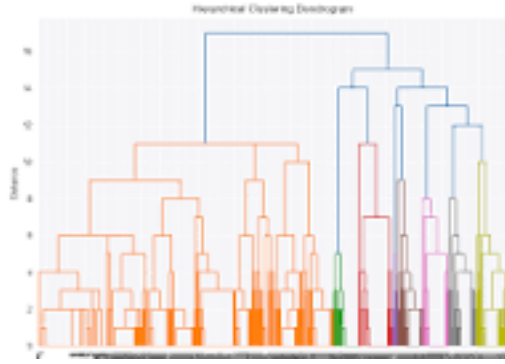


Fig. 32: Agglomerative Clustering Visualization

3) DBSCAN Clustering Visualization:

The DBSCAN visualization shows the outcome of density-based clustering, in which dense regions are merged into clusters while sparse regions are considered noise or outliers. This visualization helps you identify strange cluster shapes and find anomalous customer behavior.

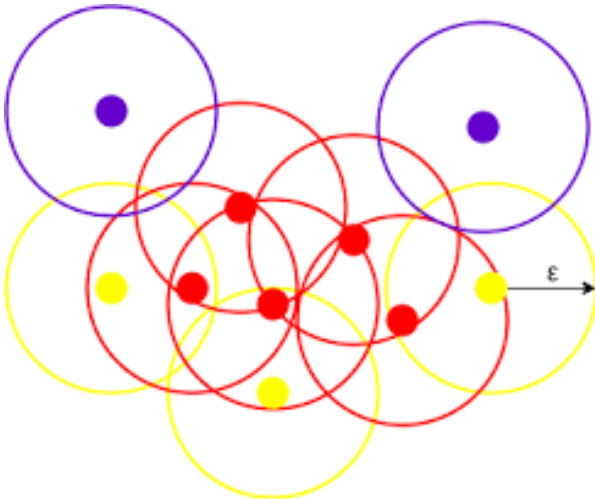


Fig. 33: DBSCAN Clustering Visualization

4) Mean Shift Clustering Visualization:

Visualization of Mean Shift clustering shows how data points tend to converge to dense areas in the feature space. The density estimation automatically detects the centers of the clusters. This allows for useful

visualizations of natural data grouping.

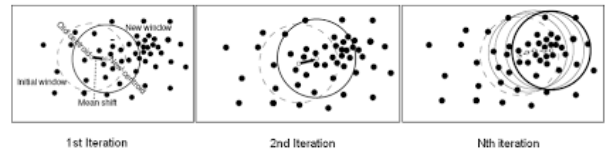


Fig. 34: Mean Shift Clustering Visualization

5) Gaussian Mixture Model Visualization:

GMM stands for Gaussian Mixture Model. It is a probabilistic clustering technique, where each point is assigned to clusters based on probability distributions. This visualization can be used to analyze the overlapping clusters and complex customer.

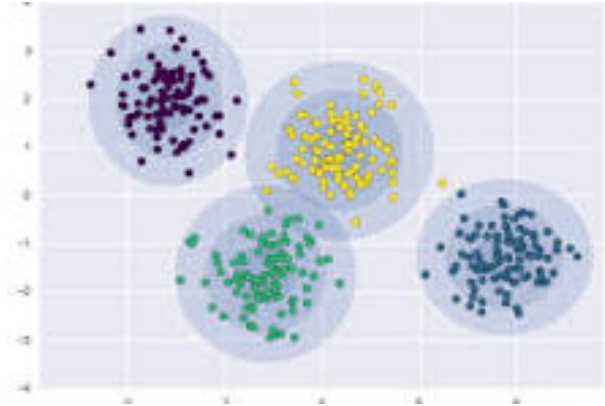


Fig. 35: Gaussian Mixture Model Visualization

6) OPTICS Clustering Visualization:

OPTICS visualization shows density-based clustering structures and ordering relations between data points. It helps to find clusters of different densities, and it offers more flexibility to discover hidden structures.

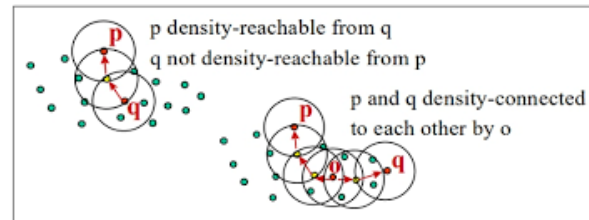


Fig. 36: OPTICS Clustering Visualization

IV. RESULTS AND EVALUATION

The algorithms used in this study were:

- K-Means Clustering

- Agglomerative Clustering
- DBSCAN
- Mean Shift
- Gaussian Mixture Model (GMM)
- OPTICS

A. Results for the First data set

For the first Dataset the results was:

TABLE VI: Clustering Algorithms Performance Evaluation

ID	Algorithm	Clusters	Silhouette	Davies-Bouldin	Calinski-Harabasz
0	K-Means	5	0.2465	1.6663	1250.4228
1	Agglomerative	5	0.2465	1.6663	1250.4228
2	DBSCAN	5	0.2320	2.5899	969.8234
3	Mean Shift	1	NaN	NaN	NaN
4	Gaussian Mixture Model	5	0.2465	1.6663	1250.4228
5	OPTICS	5	0.2448	1.5863	1002.4624

The experimental results showed that K-Means, Agglomerative Clustering, and Gaussian Mixture Model achieved the best overall performance among the evaluated algorithms[3, 2][18]. These algorithms produced identical evaluation scores, indicating that the dataset naturally forms relatively stable customer groups when centroid-based and probabilistic clustering methods are applied.

The Silhouette Score values obtained by these algorithms were approximately 0.2465, indicating moderate cluster separation. This suggests that the customer segments partially overlap, which is considered normal in customer behavior datasets due to similarities in purchasing patterns, spending habits, and online activity [4].

Additionally, the Calinski-Harabasz Score achieved by these algorithms was relatively high, reaching approximately 1250.4228. This indicates that the generated clusters maintained acceptable compactness and separation between customer groups.

DBSCAN achieved lower performance compared to the centroid-based algorithms [3]. The lower Silhouette Score and higher Davies-Bouldin Score indicate that density-based clustering was less effective for this dataset. This behavior occurred because the customer data distribution contains overlapping behavioral patterns and varying density levels, making density-based separation more difficult[20].

Mean Shift clustering failed to generate multiple clusters and classified all data points into a single cluster [?]. This result suggests that the dataset does not contain strong density peaks required by the Mean Shift algorithm.

OPTICS achieved relatively stable results and slightly improved the Davies-Bouldin Score compared to the other algorithms. This indicates that OPTICS was more capable of handling local density variations inside the dataset while maintaining acceptable cluster separation.

Overall, the experimental analysis demonstrated that centroid-based and probabilistic clustering approaches were more suitable for the customer segmentation dataset compared to density-based methods [1] [2] [3]. The results also confirmed that the dataset contains moderately separable customer groups with overlapping behavioral characteristics, which reflects realistic customer behavior patterns in retail systems.

For the Visualization:

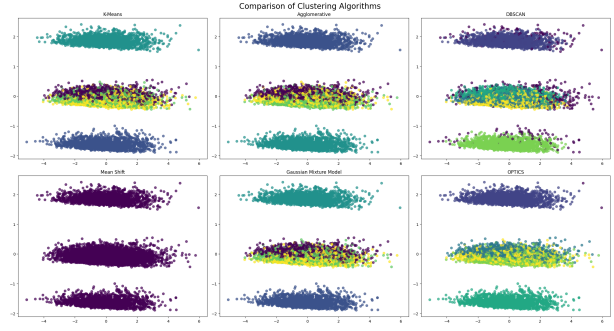


Fig. 37: First Data virtualization

Here it is also a graph for the 3 evaluation scores: Silhouette, Davies-Bouldin, Calinski-Harabasz

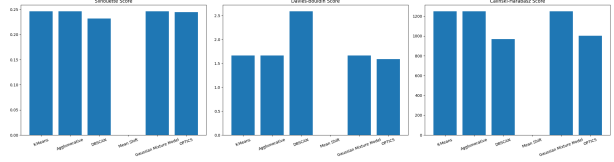


Fig. 38: First Data virtualization

And also a graph for the Best Algorithm result (K-Means) for the heatmap graph

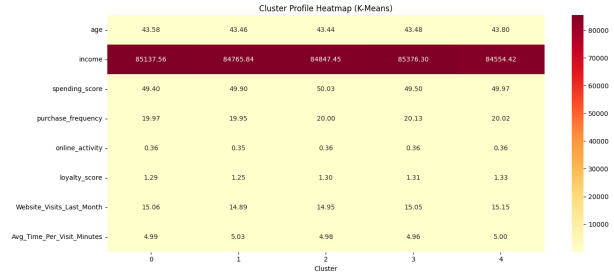


Fig. 39: First Data virtualization

B. Results for the Second data set

For the second Dataset the results was:

TABLE VII: Clustering Algorithms Performance Evaluation

ID	Algorithm	Clusters	Silhouette	Davies-Bouldin	Calinski-Harabasz
0	K-Means	4	0.2895	1.4911	2223.30
1	Agglomerative	4	0.2895	1.4911	2223.30
2	DBSCAN	4	0.2868	1.9908	1667.93
3	Mean Shift	1	NaN	NaN	NaN
4	Gaussian Mixture Model	4	0.2895	1.4911	2223.30
5	OPTICS	4	0.2895	1.4911	2223.30

K-Means, Agglomerative Hierarchical Clustering, and Gaussian Mixture Models produced highly consistent results, all identifying four distinct customer segments with nearly identical evaluation scores across multiple internal validation metrics. This strong agreement across fundamentally different algorithms indicates the presence of four robust and

natural customer groupings within the dataset, regardless of the underlying clustering approach. The Silhouette Score of 0.2895, while moderate, is typical for real-world customer behavior datasets where purchasing patterns, spending habits, and engagement levels often overlap rather than form perfectly separable groups. Nevertheless, the Calinski-Harabasz Score of 2223.30 reflects that the clusters are reasonably compact, well-separated, and exhibit good internal cohesion. These findings reinforce the stability of the four-segment solution. Furthermore, the consistency across algorithms increases confidence in the reliability of the discovered customer segments for downstream business applications. In contrast, DBSCAN achieved comparatively lower performance, recording a reduced Silhouette Score and a higher Davies-Bouldin Index, suggesting it faced challenges in handling the relatively uniform density distribution of the data. Mean Shift clustering performed poorly, collapsing the entire dataset into a single cluster, which implies the absence of prominent density peaks. On the other hand, OPTICS produced results closely aligned with the centroid-based methods, demonstrating better capability in managing varying local densities. Ultimately, K-Means was selected as the final clustering model due to its consistent performance across metrics, computational efficiency, ease of implementation, and superior interpretability. Its simplicity also allows business stakeholders to easily understand and act upon the resulting segments. Additionally, K-Means offers faster training time and better scalability, making it more practical for deployment in real-world e-commerce environments.[2, 19].

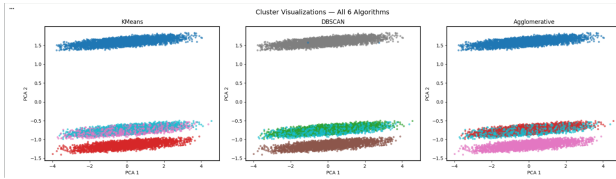


Fig. 40: Second Dataset Cluster Visualization

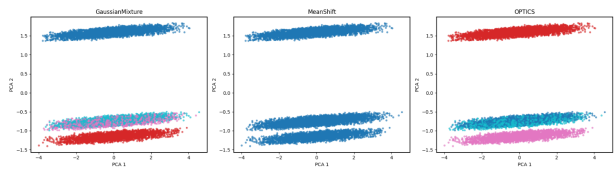


Fig. 41: Second Dataset Cluster Visualization

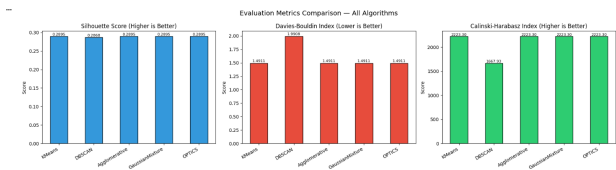


Fig. 42: Second Dataset Cluster Visualization

C. Results for the third data set

For the third Dataset the results was:

ID	Algorithm	Clusters	Silhouette	Davies-Bouldin	Calinski-Harabasz
0	K-Means	4	0.1194	2.2058	1689.1031
1	DBSCAN	3	0.0633	7.6666	651.0953
2	Agglomerative	4	0.0753	2.4190	1216.1850
3	Gaussian Mixture	4	0.0896	2.2235	1305.2612
4	Mean Shift	1	NaN	NaN	NaN
5	OPTICS	3	0.0612	13.0097	633.7920

The experimental results showed that K-Means achieved the best performance among all evaluated algorithms, obtaining the highest Silhouette Score of 0.1194 and the lowest Davies-Bouldin Score of 2.2058.[11, 12, 13] This means K-Means produced the most compact and well-separated clusters across the dataset.

The Silhouette Score values obtained across all algorithms were generally low, ranging between 0.06 and 0.12. This means customer segments in this dataset partially overlap, which is considered normal in real-world shopping behavior datasets due to gradual transitions in purchasing patterns, income levels, and digital engagement activities rather than sharp behavioral boundaries between customer groups.

Gaussian Mixture Model achieved the second-best performance with a Silhouette Score of 0.0896 and a Davies-Bouldin Score of 2.2235, which is nearly identical to K-Means. This means the dataset contains customer groups with soft overlapping boundaries that models can also capture effectively.

Agglomerative Clustering achieved moderate results with a Silhouette Score of 0.0753 and a Davies-Bouldin Score of 2.4190. Although the results are close to K-Means, the slightly higher Davies-Bouldin Score suggests that the bottom-up merging strategy of Agglomerative Clustering introduced marginally more overlap between clusters, likely due to the large size of the dataset.

DBSCAN achieved lower performance compared to centroid-based algorithms, producing a Silhouette Score of 0.0633 and a notably high Davies-Bouldin Score of 7.6666.[11, 13] The algorithm also identified 104 noise points and found only 3 clusters[17]. This behavior occurred because the consumer shopping dataset contains uniformly distributed behavioral patterns with no clear density boundaries, making density-based separation less effective for this type of data.

Mean Shift clustering failed to generate multiple clusters and classified all data points into a single cluster. This result confirms that the dataset doesn't contain strong isolated density peaks required by the Mean Shift algorithm, as the customer behavioral features are continuously and evenly distributed across the feature space.

OPTICS produced the weakest results among all evaluated algorithms, Silhouette Score of 0.0612 and the highest Davies Bouldin Score of 13.0097 across only 3 clusters. Similar to DBSCAN, OPTICS relies on local density variations to form clusters, and the uniform distribution of customer behavioral attributes in this dataset made it unable to identify meaningful and well-separated customer groups.

Overall, the experimental analysis demonstrated that centroid-based and clustering approaches were significantly more suitable for the Consumer Shopping Trends dataset compared to

density-based methods. K-Means was selected as the optimal algorithm based on its superior Silhouette and Davies-Bouldin scores [2] [3],[11, 12, 15] and was used to generate the final four customer segments: Loyal Frequent Buyers, Budget Casual Shoppers, Premium Power Shoppers, and Wealthy Occasional Buyers.

For the Visualization:

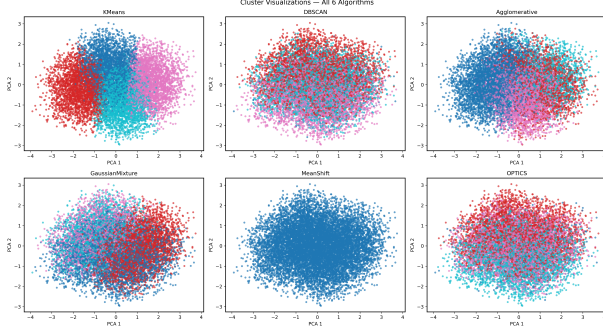


Fig. 43: Third Data virtualization

D. Results for the Fourth data set

For the fourth dataset, the results were:

TABLE VIII: Clustering Algorithms Performance Evaluation

ID	Algorithm	Clusters	Silhouette	Davies-Bouldin	Calinski-Harabasz
0	K-Means	4	0.1534	1.9623	18341.7
1	DBSCAN	6	0.0603	2.8816	3219.4
2	Agglomerative	4	0.1248	2.1917	14275.2
3	Gaussian Mixture	4	0.1090	1.8223	11998.7
4	Mean Shift	1	Nan	Nan	Nan
5	OPTICS	3	0.0164	2.7962	7079.4

The results showed that K-Means worked best on the fourth dataset, outperforming all the other clustering methods tested. It sorted customers into four groups and earned the highest Silhouette Score, which means it did the best job of clearly separating different types of customers.

K-Means achieved a Silhouette Score of about 0.1534, showing that the customer groups are only moderately distinct from one another. This means there's some overlap between the groups, which makes sense for customer behavior data—after all, many customers have similar spending habits, incomes, and shopping patterns.

K-Means got the highest Calinski-Harabasz Score—about 18,341.7—which means the groups it found were tight and still clearly different from each other. In other words, it did a great job of sorting customers into well-defined groups.

Agglomerative Clustering also grouped customers into four clusters, with a Silhouette Score of approximately 0.1248. Although its performance was lower than K-Means, the results show that hierarchical clustering was still capable of identifying meaningful customer groups. The slightly lower score suggests that some customer segments were not fully separated. This behavior occurred because Agglomerative Clustering combines customers progressively based on

distance similarity, which may merge customers with similar purchasing behavior into overlapping groups.

The Davies-Bouldin Index obtained by Agglomerative Clustering was approximately 2.1917, which was higher than K-Means. This indicates that the clusters were less separated and shared similar characteristics such as income, loyalty level, and online activity, reducing the distinction between some customer segments.

Gaussian Mixture Model also generated four clusters but achieved lower performance compared to K-Means. The Silhouette Score reached approximately 0.1090, indicating higher overlap between customer groups. This occurred because Gaussian Mixture assigns customers to clusters using probability distributions rather than strict boundaries, allowing customers with mixed behavioral patterns to partially belong to multiple clusters.

Despite the lower Silhouette Score, Gaussian Mixture achieved the lowest Davies-Bouldin Index among the algorithms that generated multiple clusters, approximately 1.8223. This indicates that the clusters were relatively compact internally. The model was effective in handling customers whose behavior patterns overlapped across multiple segments.

However, the Calinski-Harabasz Score for Gaussian Mixture was lower than both K-Means and Agglomerative Clustering, reaching approximately 11998.7. This indicates that although the clusters were internally compact, the separation between different customer groups remained weaker than K-Means.

DBSCAN produced six clusters but achieved lower clustering quality. The low Silhouette Score and high Davies-Bouldin Index indicate weak cluster separation. This suggests that density-based clustering was less suitable for this customer dataset due to overlapping behavioral patterns and varying customer densities.

Mean Shift failed to identify multiple clusters and classified all customers into a single cluster. This result indicates that the dataset does not contain strong density peaks required by the Mean Shift algorithm.

OPTICS produced three clusters but showed the lowest overall clustering quality among valid multi-cluster methods. The very low Silhouette Score indicates that customer boundaries were highly overlapping and less distinguishable.

Overall, the experiments showed that K-Means was the best method for grouping customers in the fourth dataset. The results suggest that while customer groups aren't completely separate, K-Means did a better job than other methods because it works well with the smooth, continuous patterns found in how people shop [4]. This means that, for this kind of customer data, K-Means is the most effective way to find useful and understandable customer segments[19].

For the Visualization:

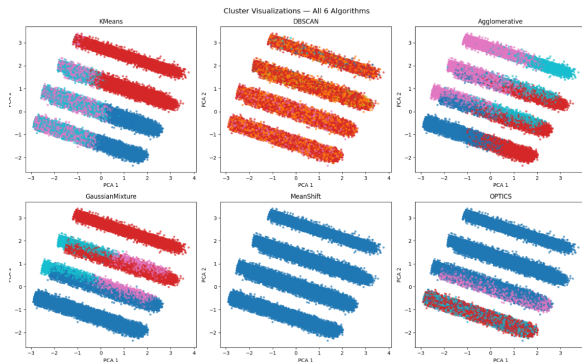


Fig. 44: Fourth Data virtualization

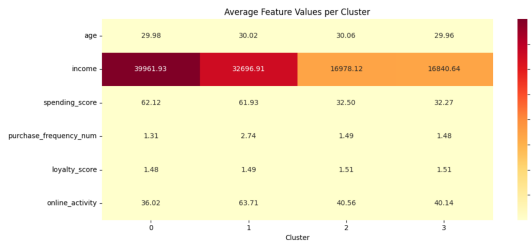


Fig. 45: Fourth Data virtualization

Here's a heatmap that shows the results from the best-performing algorithm, K-Means. This visualization highlights the differences between the customer groups found by K-Means, making it easier to spot patterns and compare their characteristics.

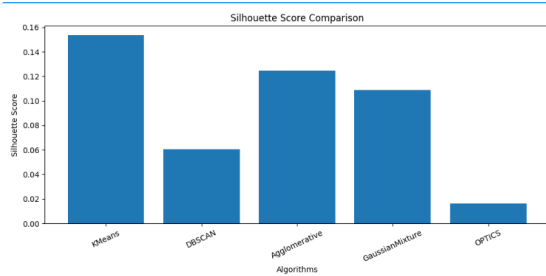


Fig. 46: Fourth Data virtualization

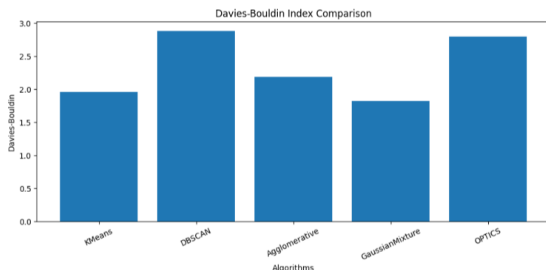


Fig. 47: Fourth Data virtualization

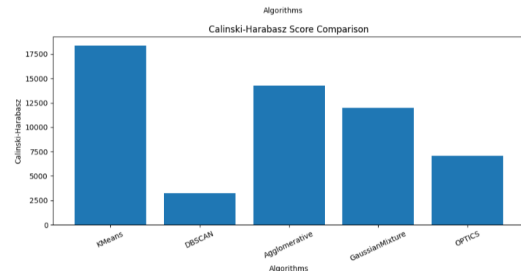


Fig. 48: Fourth Data virtualization

Here's a graph that compares all three evaluation scores—Silhouette, Davies-Bouldin, and Calinski-Harabasz—for the different clustering methods. This makes it easy to see how each method performed across the key metrics.

V. CONCLUSION

Results and Discussion Customer segmentation plays a vital role in modern retail and e-commerce, enabling businesses to better understand diverse customer behaviors and tailor strategies to their specific needs. In this study, six unsupervised clustering algorithms — K-Means, Agglomerative Clustering, DBSCAN, Mean Shift, Gaussian Mixture Models, and OPTICS — were comprehensively evaluated across four real-world datasets of varying sizes (ranging from approximately 9,000 to 100,000 records, with some downsampled for consistency). Three composite features (Spending Score, Loyalty Score, and Online Activity) were carefully engineered from underlying attributes to capture richer behavioral signals and improve clustering quality. Across all datasets, centroid-based and probabilistic algorithms consistently outperformed density-based methods. On the Online Retail Customer dataset (9,000 records), K-Means, Agglomerative Clustering, and Gaussian Mixture Models achieved the highest Silhouette Score of 0.2465, with OPTICS also performing competitively, while Mean Shift failed to detect multiple meaningful clusters. Similar strong performance was observed on the second dataset (10,000 records), yielding a Silhouette Score of 0.2895, where K-Means was ultimately selected for its balance of effectiveness, scalability, and interpretability. For the Consumer Shopping Trends Dataset (12,000 records), K-Means delivered the best results, successfully identifying four realistic and actionable customer personas: Loyal Frequent Buyers, Budget Casual Shoppers, Premium Power Shoppers, and Wealthy Occasional Buyers. On the fourth dataset (Customer Purchases Behaviour, downsampled to 10,000 records), K-Means again led with a Silhouette Score of 0.1534. The obtained Silhouette Scores ranged moderately from 0.06 to 0.29 across the four datasets. These moderate values are reasonable and expected for real-world customer data, as consumer buying behaviors are often distributed gradually rather than forming sharply distinct groups. Overall, K-Means emerged as the most robust, consistent, and practical performer across different dataset sizes and characteristics. These findings demonstrate that

well-engineered composite features combined with centroid-based clustering can effectively uncover meaningful customer segments even in complex retail environments. The identified personas offer clear business value, allowing for targeted marketing, personalized promotions, inventory planning, and customer retention strategies tailored to each group's unique behavior. Future Work In future studies, we plan to incorporate temporal components such as purchase recency and seasonality to further enhance clustering effectiveness.

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